UTKARSH JOSHI

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OBJECTIVE

Systems programmer working on simulation, numerical computing and the machine code underneath both. Started with Linux to fix a hardware problem in school, which led to C, C++, CUDA and OpenGL. Interested in software engineering internships where the work is measured.

SKILLS

Languages C, C++, Python, Bash **Systems** Linux/WSL2, Git, CMake, Docker, OpenMP, MPI, cgroups, perf_event_open **Libraries** CUDA, CuPy, NumPy, SciPy, PySide6, Textual, FastAPI, pybind11, ctypes

SELECTED WORK

S-Stream **lattice Boltzmann fluid workbench — Python, CuPy, PySide6 · on PyPI**

- D2Q9 BGK solver with bounce-back boundaries and Shan-Chen multiphase; the whole lattice runs on the GPU through a CuPy path.
- Obstacles are drawn interactively, probes are dropped into the flow and read live, and parameter sweeps run in batch.
- Every running simulation carries a confidence score, and capabilities are labelled Verified, Experimental or Hidden in TRUST.md. Validated against Poiseuille flow, a lid-driven cavity and measured cylinder drag coefficients — the tests are in the repository.

noether **symplectic integration on GPUs — C++20, CUDA · with a paper**

- Measures Hamiltonian energy drift and reduction non-determinism separately, because they are different failures and a GPU makes both.
- Reductions are run a hundred times over n = 2^20 elements for every mode and precision combination, then compared bit for bit. Mixed precision runs FP32 with Kahan compensation on the kick against an IEEE-conformant FP64 baseline with no fast-math anywhere in the build.
- Four-page paper with the figures and the raw result CSVs are in the repository.

Visage **live machine dashboard — Python, Textual, perf_event_open, eBPF**

- CPU, memory, disk, network, GPU and processes, sampled from psutil and /proc with GPU numbers from NVML.
- The benchmark path is the interesting one: cores pinned with taskset, CPU frequency locked, and PMU counters read straight out of perf_event_open through ctypes. Every result carries the system context that produced it.
- Remote monitoring over FastAPI; results export as JSON or CSV.

CTorrent **BitTorrent client — C11, POSIX sockets · no third-party libraries**

- The constraint is the point: how small can a working client be when the only dependencies are the C standard library and POSIX sockets? About 2,500 lines.
- Bencoded metainfo, the info hash, the wire messages, rarest-first selection and piece verification — HTTP and HTTPS trackers, the full handshake, bitfield, have, choke, request and piece messages, 16 KiB blocks with 16 in flight per peer, SHA-1 verification of every piece.
- Not fast. Legible.

quip **Unix shell written from scratch — C99, termios, POSIX**

- Raw terminal mode by hand, pipelines, redirection, job control, signal handling, tab completion, history on disk.
- Runs interactively and non-interactively, survives unmatched quotes and trailing backslashes, and tracks background jobs properly. Plugins are Python files loaded at startup and called as commands.

Simulation Zoo **thirteen deterministic simulations — C++20, headless renderer**

- A self-gravitating galaxy, Gray-Scott coral, Lenia blooms, Nagel-Schreckenberg traffic, an Abelian sandpile — each with exactly one signature visual.
- Fixed-timestep, seeded and reproducible: the same run always produces the same frames, headless, with no display and no GPU.

BadType **a very very horrid language — C99 type checker and REPL**

- Seven type rules, none of them about values — they are about structure, order, scope and time. Addition returns a Mood, which then behaves perfectly legally everywhere a Mood is accepted.
- The compiler is technically right and completely insane. Every diagnostic is hand-written. No dependencies, no CI, no badges.

Versus **five spatial-indexing engines race — C11, KD-tree, grid, octree, BVH**

- Central-gravity orbits and close approaches, to compare how fast each index finds nearby pairs of moving objects without checking every pair against every other.
- Both panels advance independently, so the faster engine completes more steps while the slower one is still working; benchmark mode pins both to the same fixed step count when a fair measurement is wanted.

seams **catalogue of where compute meets memory — Python, CUDA, MPI**

- Ten failure seams, each with the theory of why it tears, a plot that exposes it, the smallest program that provokes it, and the remedy.
- Organised by failure mode rather than technique, and every seam is labelled hands-on, emulated or theory, so it is always clear whether the pathology was reproduced on real hardware or merely argued about.

WarpX **Particle-In-Cell simulation — C++, AMReX, CUDA, MPI · fork**

- A fork of our own upstream rather than of the canonical repository, so the tree is ours to work in: read it, break it, push patches back.

PressPNG **compress an image to a required target size — C++17, Win32, CMake**

- The target is a size, not a quality knob: choose the resolution, then the quantisation, then the colour depth, in an order that converges.
- Native Win32 GUI and a CLI over the same engine, no runtime dependencies, Google Test for the cases that matter.

EDUCATION

Graphic Era Hill University, Bhimtal, Uttarakhand
B.Tech, Computer Science and Engineering · SGPA 8.71 · expected May 2029

ACHIEVEMENTS

- Top 10 Finalist, National Space Hackathon 2026, organised by IIT Delhi.
- Self-study: MIT *Missing Semester of Computer Science* completed; currently reading CS:APP.